

Grassmannian manifold

Visual intuition, graduate-level mathematics, and a data-association case study.

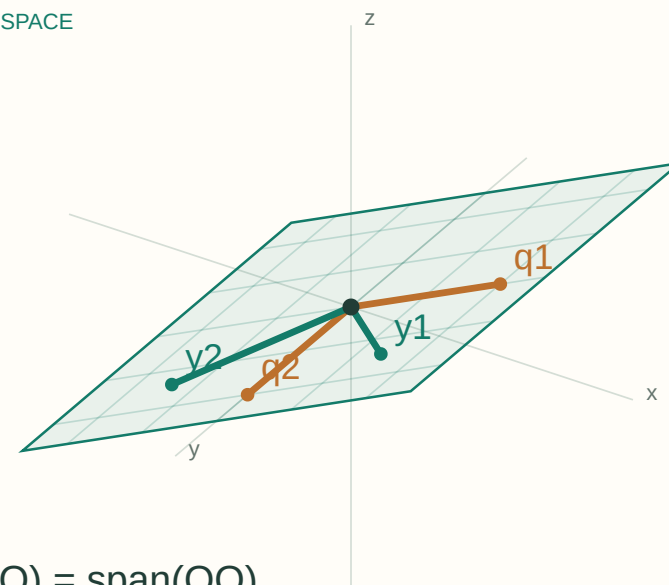
A space of subspaces.

When a line, plane, or learned subspace becomes a single geometric point.

34 SLIDES / FIVE LIVE EXPERIMENTS

Includes Lusk & How (2022):
global association of 3-D lines and planes.

ONE SUBSPACE



$$\text{span}(Q) = \text{span}(QO)$$

[Companion guide & live labs ↗](#)

From geometry to global matching.

Choose a chapter, or use the arrow keys to follow the story.

01 The subspace is the point →

Bases, projectors, and quotient geometry

02 Measure a difference →

Principal angles and distance conventions

03 Move without leaving →

Tangents, geodesics, logarithms, retractions

04 Learn a subspace →

PCA as a Grassmann optimization problem

05 Match lines and planes →

The requested paper and a synthetic demo

06 Keep the distinctions →

A formula sheet and primary references

A point can represent an entire plane.

Fix the dimension k . Let the subspace vary.

THE REAL GRASSMANNIAN

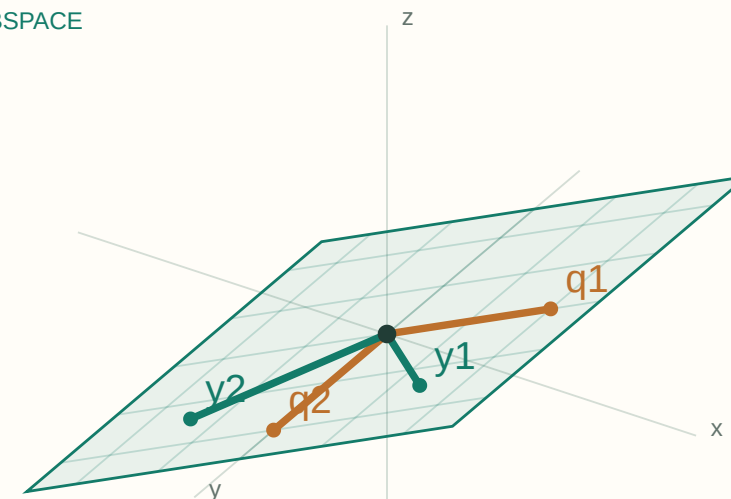
$$\text{Gr}(k, n) = \{\mathcal{S} \subset \mathbb{R}^n : \dim \mathcal{S} = k\}$$

A point is a **linear subspace**, not an individual vector in it.

All subspaces contain the origin.

Our notation: k first, ambient n second. We assume $0 < k < n$.

ONE SUBSPACE



$$\text{span}(Q) = \text{span}(QO)$$

Change the basis. Keep the point.

The coordinates inside a subspace are not part of its identity.

AN ENTIRE EQUIVALENCE CLASS

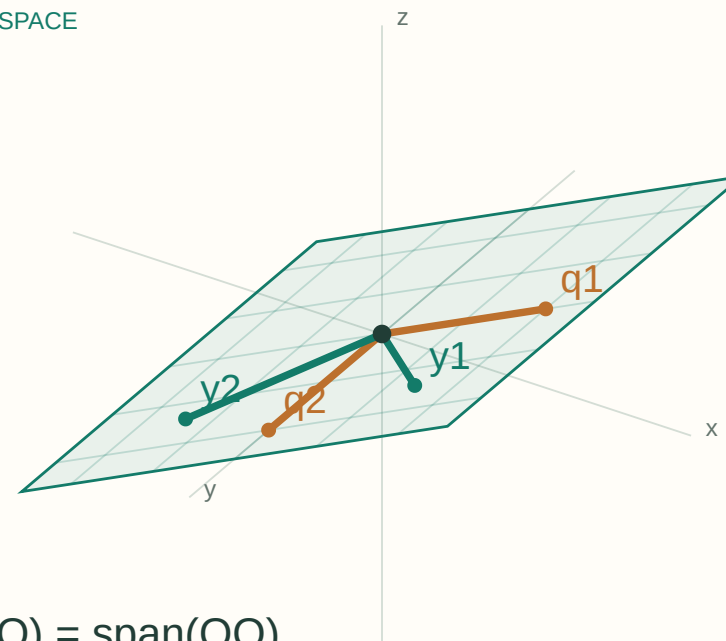
$$Q^T Q = I_k, \quad [Q] = \{QO : O \in O(k)\}$$

Rotations, reflections, and column sign flips preserve the column space.

In the lab, rotate the arrows and watch the plane remain fixed.

Static example: a 90° basis rotation gives $\|Q - QO\|_F = 2$; the subspace distance is 0.

ONE SUBSPACE

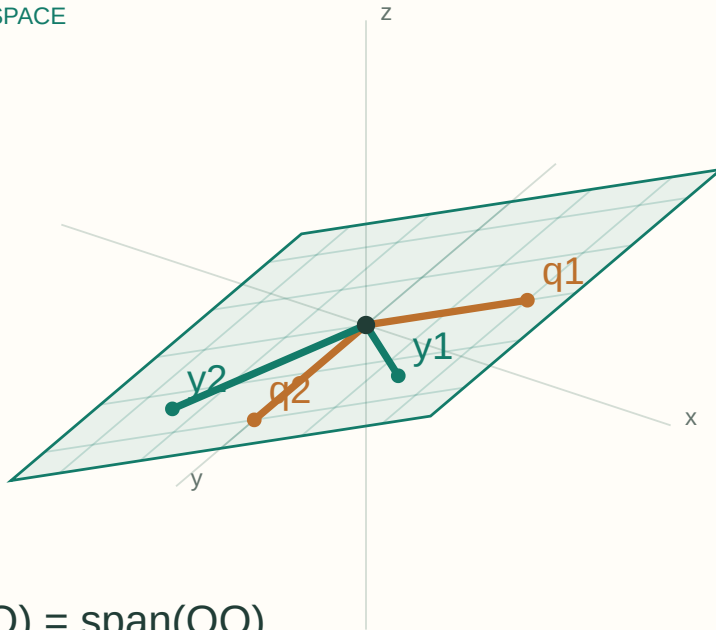


$$\text{span}(Q) = \text{span}(QO)$$

One plane. Many bases.

Rotate the basis, then tilt the plane. Watch which quantities change.

ONE SUBSPACE



$$\text{span}(Q) = \text{span}(QO)$$

Quarter turn → same plane.

ROTATE, THEN TILT

A quarter turn changes the basis matrix: $\|Q - QO\|_F = 2$. The projector difference stays zero.

Tilting the plane changes the subspace itself.

Remove the degrees of freedom of the basis.

Stiefel keeps a frame. Grassmann keeps only its span.

THE MATHEMATICAL STATEMENT

$$\text{Gr}(k, n) = \text{St}(k, n) / O(k), \quad \dim \text{Gr}(k, n) = k(n - k)$$

COUNT THE PARAMETERS

$$nk - \frac{k(k+1)}{2} - \frac{k(k-1)}{2}$$

Orthonormality removes $k(k+1)/2$ parameters; internal rotations remove $k(k-1)/2$.

READ THE QUOTIENT

Each orthonormal frame belongs to a whole family QO.

The quotient identifies that family with one subspace.
Gr(2,3) has dimension 2.

The projector forgets the basis automatically.

Every k -dimensional subspace has exactly one orthogonal projector.

THE MATHEMATICAL STATEMENT

$$P = QQ^\top = (QO)(QO)^\top, \quad P^2 = P = P^\top, \quad \text{tr } P = k$$

WHAT P DOES

$$\hat{x} = Px, \quad x - Px \perp \text{span}(Q)$$

Px is the least-squares approximation of x within the subspace.

WHAT IT COSTS

Q stores nk entries; P stores n^2 entries.

P is unique, but an orthonormal basis is usually cheaper computationally.

Lines identify opposite directions.

A sphere picture needs an antipodal identification.

$\text{Gr}(1,2)$ AND $\text{Gr}(1,3)$

$$[u] = [-u], \quad \|u\| = 1$$

$\text{Gr}(1,2)$: unoriented lines in a plane.

$\text{Gr}(1,3) \cong \mathbb{RP}^2$: sphere points u and $-u$ represent the same line.

$\text{Gr}(2,3)$

A plane through the origin is determined by its normal direction.

Normals n and $-n$ describe the same plane.

Offset planes need affine geometry.

Matrix distance can report a change that never happened.

Two representatives may look far apart while the subspace is identical.

COMPARE Q AND $-Q$

$$\|Q - (-Q)\|_F = 2\sqrt{k}$$

$$QQ^\top = (-Q)(-Q)^\top$$

Basis distance is sensitive to signs and rotations.

COMPARE THE SUBSPACES

$$d_g([Q], [-Q]) = 0$$

Use principal angles or projector distances.

Never report raw basis error as subspace error without alignment.

Find directions that align best.

The singular values of $Q^T Y$ remove both choices of basis.

SVD → ANGLES

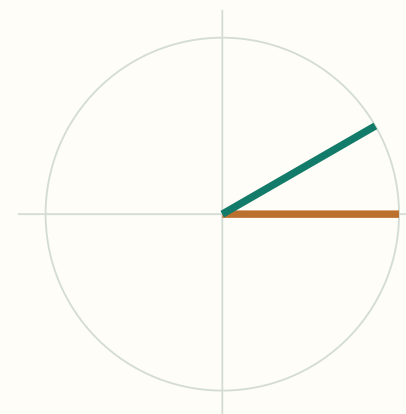
$$Q^T Y = U \operatorname{diag}(\sigma_i) V^T$$

$$\theta_i = \arccos \sigma_i$$

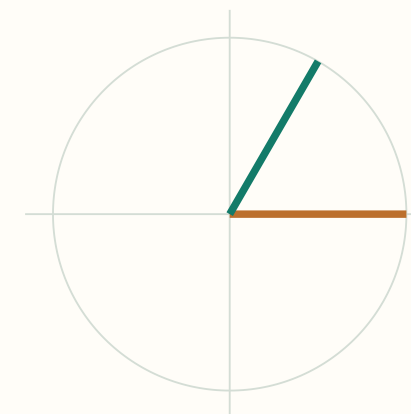
$$0 \leq \theta_1 \leq \dots \leq \theta_k \leq \pi/2$$

The corresponding principal vectors are columns of QU and YV .

(e_1, e_3) slice



(e_2, e_4) slice



Two independent rotations in $\mathbb{R}^4$

Two useful distances. Two different geometries.

The normalization and angle units belong in the formula.

INTRINSIC AND PROJECTOR DISTANCE

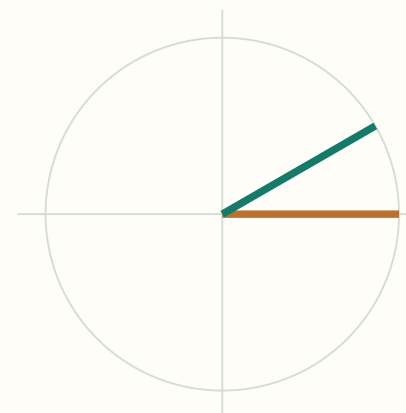
$$d_g = \sqrt{\sum_i \theta_i^2}$$

$$d_p = \frac{\|QQ^\top - YY^\top\|_F}{\sqrt{2}}$$

$$d_p = \sqrt{\sum_i \sin^2 \theta_i}$$

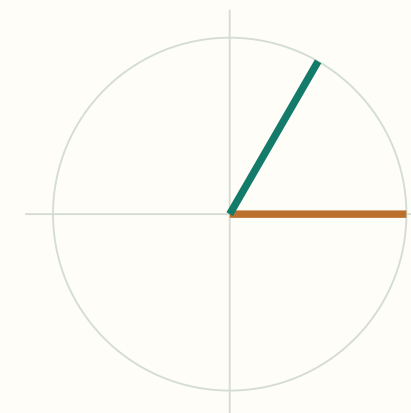
Angles in **radians**. For small angles, $\sin \theta \approx \theta$; globally these distances differ.

(e₁, e₃) slice



30°

(e₂, e₄) slice



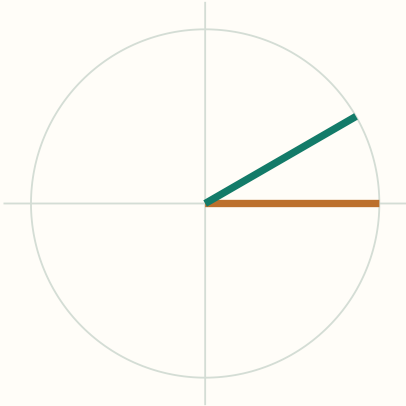
60°

Two independent rotations in $\mathbb{R}^4$

Principal angles and distances

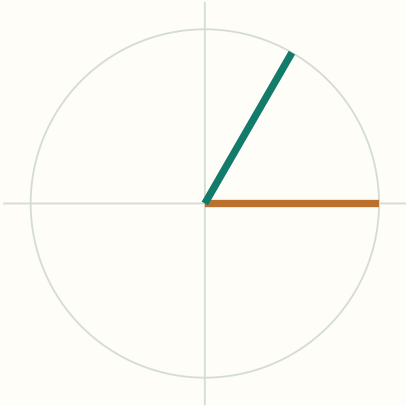
Change the two coordinate rotations; then rotate the target basis.

(e₁, e₃) slice



30°

(e₂, e₄) slice



60°

Two independent rotations in R⁴

Two independent angles in Gr(2,4).

COMPARE THE GEOMETRIES

At 30° and 60°, the geodesic distance is $\pi\sqrt{5}/6 \approx 1.171$ rad; the projector distance is 1.

Rotating the basis changes neither distance.

Two planes in $\mathbb{R}^3$ must share a direction.

The experiment uses $\mathbb{R}^4$ so that both principal angles can vary.

THE MATHEMATICAL STATEMENT

$$Q = [e_1, e_2], \quad Y = [\cos \alpha e_1 + \sin \alpha e_3, \cos \beta e_2 + \sin \beta e_4]$$

IN FOUR DIMENSIONS

$$(\alpha, \beta) = (30^\circ, 60^\circ)$$

$$d_g = \pi\sqrt{5}/6 \approx 1.171, \quad d_p = 1$$

IN THREE DIMENSIONS

$$\dim(S \cap T) \geq \dim S + \dim T - 3 = 1$$

For two planes, $\theta_1 = 0$; only θ_2 varies.

A tangent points out of the current subspace.

Horizontal representatives remove motion that only changes the basis.

TANGENT IN BASIS COORDINATES

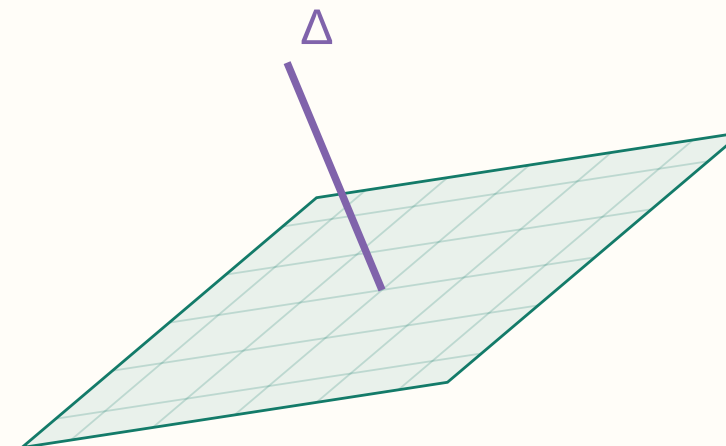
$$Q^\top \Delta = 0$$

$$\Pi_Q(Z) = (I - QQ^\top)Z$$

Internal motion $Q\Omega$, with $\Omega^\top = -\Omega$, changes the frame alone.

A horizontal Δ changes the Grassmannian point.

LEAVE THE SUBSPACE



$$Q^\top \Delta = 0$$

Rotate the active directions with sines and cosines.

For a compact rank- r SVD $\Delta = U\Sigma V^\top$, use the canonical metric.

THE MATHEMATICAL STATEMENT

$$Q(t) = Q + QV(\cos(t\Sigma) - I_r)V^\top + U\sin(t\Sigma)V^\top$$

WHAT THE FORMULA GUARANTEES

$$Q(0) = Q, \quad \dot{Q}(0) = \Delta$$

$$Q(t)^\top Q(t) = I_k$$

Zero singular directions stay fixed.

LENGTH IS NOT ALWAYS DISTANCE

The speed is $\|\Delta\|_F$.

A sufficiently long geodesic is no longer a shortest path.
Cut points matter when inverting the map.

Follow a geodesic between unoriented lines.

A great-circle arc is a representative path, with antipodal points identified.

THE $\text{Gr}(1,3)$ EXAMPLE

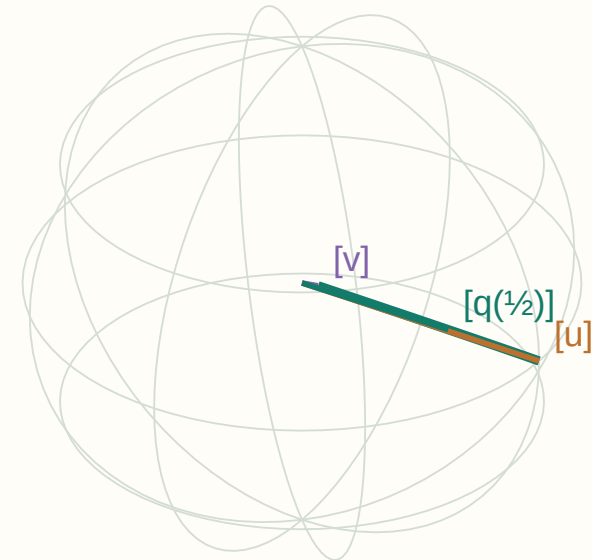
$$q(t) = u \cos(t\theta) + w \sin(t\theta)$$

$$w = \frac{v - u \cos \theta}{\sin \theta}$$

First choose the sign of v so $u^T v \geq 0$.

For $0 \leq t \leq 1$: distance from the start = $t\theta$.

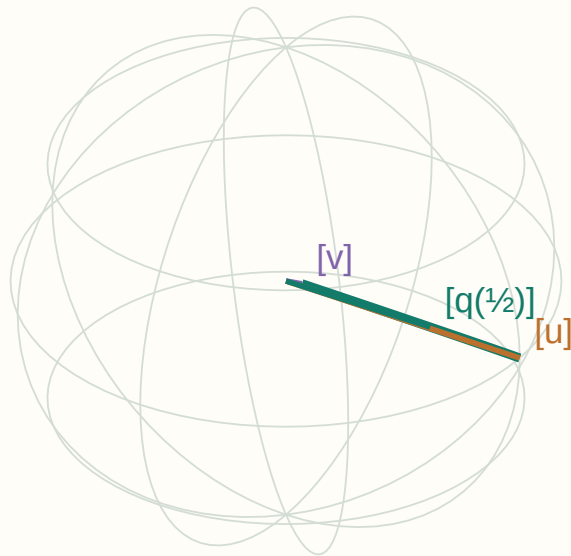
At $\theta = 0$, the path is constant.



q and $-q$ represent the same line

Follow a Grassmann geodesic

Move along the path, animate it, or flip the endpoint vector.



q and $-q$ represent the same line

The sphere is a representative picture.

FOLLOW THE SHORTEST PATH

At endpoint angle 65° and $t = 0.4$, the distance from the start is $26^\circ \approx 0.454$ rad.

The line is unchanged by an endpoint sign flip.
At 90° , shortest paths are not unique.

The logarithm is local, not globally single-valued.

A principal angle of 90° marks the boundary of a unique shortest logarithm.

AWAY FROM THE CUT LOCUS

$$Z = (I - QQ^\top)Y(Q^\top Y)^{-1}$$

$$Z = U\Sigma V^\top$$

$$\text{Log}_{[Q]}[Y] = U \arctan(\Sigma) V^\top$$

This gives a horizontal tangent representative.

AT THE BOUNDARY

If a principal angle is $\pi/2$, $Q^\top Y$ is singular.

There are multiple minimizing logarithms. Near the boundary, this inverse-based expression is numerically fragile.

Use robust algorithms from the handbook.

An orthonormalization can be a local step.

A retraction matches the exponential to first order; it is cheaper to compute.

THE MATHEMATICAL STATEMENT

$$R_Q(\Delta) = \text{qf}(Q + \Delta), \quad Q_+ = R_Q(-\eta \text{grad } f(Q))$$

PRESERVE THE CONSTRAINT

QR returns an orthonormal frame.

Column sign conventions do not change the represented subspace.

CHOOSE THE STEP

A tangent is local. Use a line search or a trust region to control the optimization step.

QR retraction is generally not the exact exponential.

Learn the subspace that retains the most variance.

C is the covariance of centered data; the representation Q is orthonormal.

THE MATHEMATICAL STATEMENT

$$f(Q) = -\text{tr}(Q^\top C Q), \quad \text{grad } f(Q) = -2(I - Q Q^\top) C Q$$

WHY GRASSMANN?

$$f(QO) = f(Q)$$

Only the span matters.

Reconstruction loss = $\text{tr}(C) - \text{tr}(Q^\top C Q)$.

WHAT IS THE OPTIMUM?

The span of the k leading eigenvectors.

The subspace is unique if $\lambda_k > \lambda_{k+1}$, even when eigenvalues inside it are repeated.

Watch a plane learn a point cloud.

The exact eigenspace solution supplies an independent reference.

A CONTROLLED OPTIMIZATION

$$\lambda(C) = (9, 4, 1), \quad k = 2$$

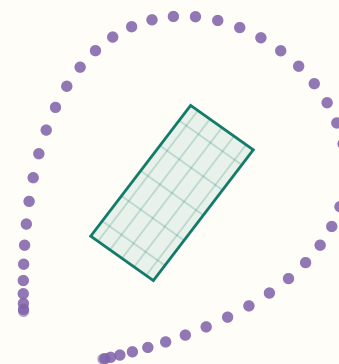
Optimum captured variance: **13**.

Optimum reconstruction loss: **1**.

Take one gradient step, run descent, or reduce λ_2 until the boundary gap closes.

The cloud covariance is computed from the plotted samples.

CAPTURE THE SHARED VARIATION

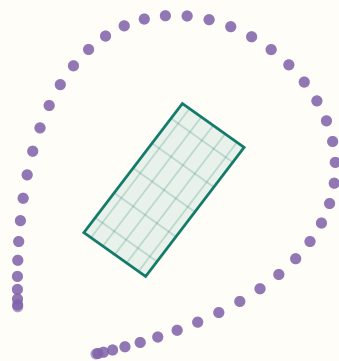


$$\lambda = (9, 4, 1) \rightarrow \text{captured variance} = 13$$

Learn a plane from a point cloud

Step through gradient descent and explore a closing eigenvalue gap.

CAPTURE THE SHARED VARIATION



$\lambda = (9, 4, 1) \rightarrow$ captured variance = 13

OPTIMIZE THE SUBSPACE

For covariance eigenvalues (9, 4, 1), the optimal plane captures variance 13 and leaves reconstruction loss 1.

When $\lambda_2 = \lambda_3$, several optimal planes exist.

The spectral solution is the reference.

A useful model whenever only a span matters.

Choose the representation that preserves the information your task needs.

REAL SUBSPACE APPLICATIONS

Subspace tracking: follow a changing signal subspace.

Reduced models: represent a low-dimensional trial space.

Recognition: compare learned feature subspaces.

COMPLEX SIGNAL SUBSPACES

$$Q^H Q = I, \quad [Q] = \{QU : U \in U(k)\}$$

$$\dim_{\mathbb{R}} \text{Gr}_{\mathbb{C}}(k, n) = 2k(n - k)$$

Use conjugate transpose and unitary changes of basis.
The labs here are real.

SLAM objects are often displaced lines and planes.

The supplied paper uses the affine Grassmannian.

ADD A DISPLACEMENT

$$\mathcal{A} = b + \text{span}(A)$$

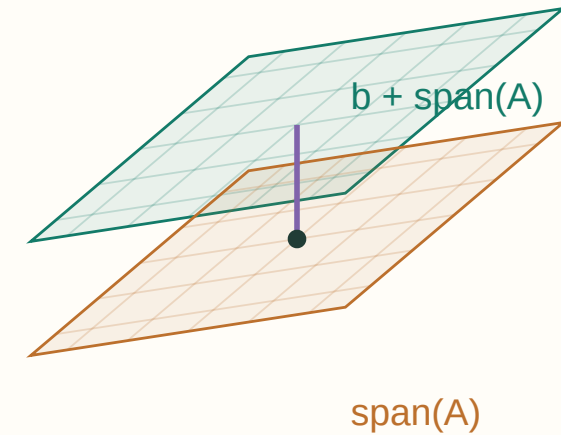
$$b_{\perp} = (I - AA^{\top})b$$

Line: $k = 1$. Plane: $k = 2$.

Moving b along the object does not change the affine subspace.

$$\dim \text{Graff}(k, n) = (k + 1)(n - k)$$

DIRECTION + ORTHOGONAL DISPLACEMENT



Affine objects need not contain the origin.

Embed an affine object in one extra dimension.

A 3-D line becomes a 2-D linear subspace in $\mathbb{R}^4$; a plane becomes a 3-D one.

ORTHONORMAL STIEFEL COORDINATES

$$Y = \begin{bmatrix} A & b_{\perp}/\eta \\ 0 & 1/\eta \end{bmatrix}$$

$$\eta = \sqrt{1 + \|b_{\perp}\|^2}$$

$$\text{Graff}(k, 3) \hookrightarrow \text{Gr}(k+1, 4)$$

Divide positions by a length scale ρ before embedding.

COMPARE EMBEDDED OBJECTS

Compute singular values of $Y_1^T Y_2$ and take the norm of the principal angles.

For mixed dimensions, retain $\min(k_1+1, k_2+1)$ angles.

Across dimensions, containment can give zero: this is a dissimilarity, not a separating metric.

Shift the pair before measuring its affine distance.

Raw homogeneous angles depend on the coordinate origin.

THE MATHEMATICAL STATEMENT

$$(b_1, b_2) \mapsto \left(0, \frac{b_2 - b_1}{\rho}\right)$$

WHY TRANSLATION CANCELS

$$(Rb_2 + t) - (Rb_1 + t) = R(b_2 - b_1)$$

Then project the shifted displacement orthogonally and form the embedding.

WHY THE SCALE MATTERS

ρ controls how positions compare with directions.

Large separations otherwise saturate angular distances.
Use the same ρ and pair order in both scans.

A vertex is a match. An edge is agreement.

Candidate $u = (i,j)$ means object A_i might correspond to object B_j .

THE MATHEMATICAL STATEMENT

$$c_{uv} = |d(A_i, A_p) - d(B_j, B_q)|, \quad M_{uv} = e^{-c_{uv}^2/(2\sigma^2)} \text{ if } c_{uv} < \epsilon$$

CANDIDATES

Match lines to lines, and planes to planes.

But an internal pair may contain a line and a plane; its consistency informs both matches.

HARD EXCLUSIONS

Set $M_{uv} = 0$ if $c \geq \epsilon$, or if two candidates reuse the same object.

Set $M_{uu} = 1$. The matrix is symmetric.

Maximize weighted density, not just match count.

A weakly connected larger clique may score worse than a smaller strong one.

THE MATHEMATICAL STATEMENT

$$\max_{u \in \{0,1\}^m \setminus \{0\}} \frac{u^\top M u}{u^\top u} \quad \text{s.t. } u_i u_j = 0 \text{ when } M_{ij} = 0$$

A SMALL EXAMPLE

Two vertices with weight 1 have density 2.

Add a third vertex with weights 0.1 to each: density drops to 1.8.

PAPER AND BROWSER

The paper uses the CLIPPER relaxation.

This small demo enumerates every feasible clique, including nonmaximal cliques, to solve the displayed discrete objective exactly.

Recover the shared objects from a scrambled scene.

Six true objects. Two distractors. No initial pose supplied to matching.

CONTROLLED MIXED-OBJECT DEMO

Scan A: 3 lines + 3 planes.

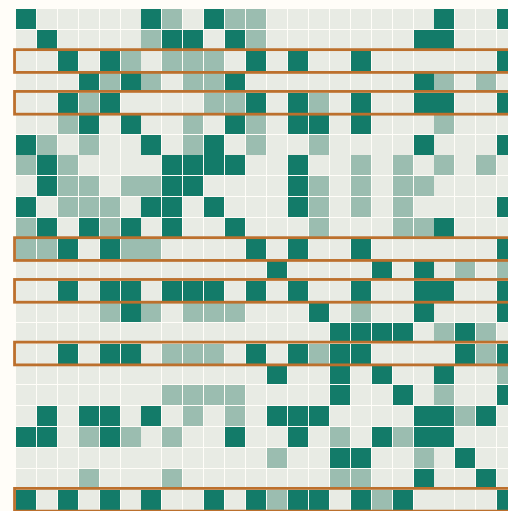
Scan B: the same objects, reordered and transformed, plus 2 distractors.

24 candidate matches.

With no noise, the 6 true matches form a unit-weight clique: density 6.

Change pose, noise, ρ , and ϵ . Compare shifted, raw, and direction-only scores.

24 × 24 CONSISTENCY MATRIX



24

candidate matches

6

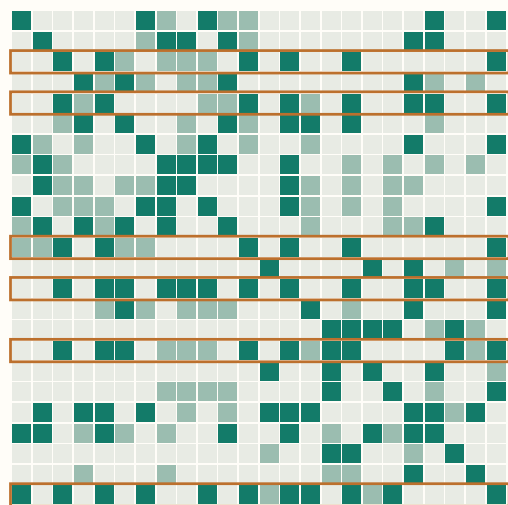
selected matches

Gold rows = one jointly consistent set

Match lines and planes

Change the scan pose, add noise, and compare the pairwise descriptors.

24 × 24 CONSISTENCY MATRIX



24

candidate matches

6

selected matches

Gold rows = one jointly consistent set

AFFINE DATA ASSOCIATION

Three lines and three planes, plus two distractors, give 24 candidate matches.

The clean shifted-affine scene recovers 6 / 6 true matches with density 6.

This small synthetic demo solves the discrete objective exactly; the paper uses CLIPPER.

Lusk & How (2022) · synthetic scene.

Association is one stage of the paper's pipeline.

A synthetic recovery result is not a loop-closure benchmark.

IMPLEMENTED IN THIS DEMO

Affine embedding; shift and scale.

Pairwise consistency and exclusions.

Exact weighted clique selection on 24 candidates.

Ground-truth recovery shown only for evaluation.

THE PAPER GOES FURTHER

Extract lines and planes from lidar.

Use CLIPPER for association.

Estimate rotation and translation from matched objects, then evaluate loop closures on KITTI.

These stages are not run in the browser.

Five distinctions keep the geometry honest.

Every one is visible somewhere in the experiments.

GEOMETRY

Basis $\neq$ subspace. Use equivalence classes.

Sphere $\neq$ Grassmannian. Identify antipodes for real lines.

Linear $\neq$ affine. Offsets need additional coordinates.

COMPUTATION

Path length $\neq$ shortest distance after a cut point.

Count $\neq$ weighted density in the association objective.

Also state radians, projector normalization, and real versus complex conventions.

A compact computational reference.

Real orthonormal bases; equal subspace dimensions unless stated otherwise.

REPRESENT · COMPARE

$$[Q] = \{QO : O \in O(k)\}, \quad P = QQ^\top$$

$$\theta_i = \arccos \sigma_i(Q^\top Y)$$

$$d_g = \|\theta\|_2, \quad d_p = \|\sin \theta\|_2$$

$$\dim \text{Gr}(k, n) = k(n - k)$$

MOVE · OPTIMIZE

$$Q^\top \Delta = 0, \quad \Pi_Q(Z) = (I - P)Z$$

$$g(\Delta_1, \Delta_2) = \text{tr}(\Delta_1^\top \Delta_2)$$

$$\text{grad } f = (I - P)\nabla f$$

$$Q_+ = \text{qf}(Q - \eta \text{grad } f)$$

Follow the formulas back to their sources.

The guide includes implementation details, numerical checks, and all five labs.

01 A Grassmann Manifold Handbook ↗

Bendokat, Zimmermann & Absil · quotient geometry, exp/log, cut locus

02 The Geometry of Algorithms with Orthogonality Constraints ↗

Edelman, Arias & Smith (1998) · §2.5 geometry; §4.5 distances

03 Global Data Association for SLAM with 3D Grassmannian Manifold Objects ↗

Lusk & How (2022) · the supplied paper; Eqs. (2)–(9), Proposition 1

[Companion guide & all experiments ↗](#)

[Authors' CLIPPER code ↗](#)

[Printable PDF ↗](#)